EasyDocs Architecture Blueprint

Spatial Workspace Models & Core Schema (Simplified Version)

This document outlines the streamlined database schema designed for the infinite spatial canvas. Business logic is deferred to the API layer to maintain lightweight, flexible data models.

1. Database Schema

Model: Board (The Infinite Canvas)		
Field	**Type**	**Details**
`id`	UUIDField	Primary Key. Secure, non-sequential routing.
`company`	ForeignKey	Cascades on deletion. Maps canvas to the tenant.
`board_type`	CharField	Choices: PRIMARY, CLIENT, CUSTOM.
`name`	CharField	Optional blank initialization.

Model: Galaxy (Visual Grouping Frame)		
Field	**Type**	**Details**
`board`	ForeignKey	Cascades on deletion.
`name`	CharField	Title of the bounded zone.
`x_position`, `y_position`	FloatField	X, Y coordinates on the Board.
`width, height`	FloatField	Dimensions of the zone.

Model: Asset (File placement)		
Field	**Type**	**Details**
`galaxy`	ForeignKey	Assets exist strictly inside a Galaxy.
`title`	CharField	Display title of the file.
`file`	FileField	Saves locally to `assets/` directory.
`x_position`, `y_position`	FloatField	X, Y relative coordinates.
`uploaded_by`	ForeignKey	Uses `SET_NULL`. Retains company files if user leaves.

2. REST API Flow

Because the models are purely structural, the API handles all constraint enforcement. The endpoints drive the React frontend's infinite canvas state.

GET `/api/workspace/board/primary/`

Purpose: Fetches the primary board object, nested with its child Galaxies and their respective Assets in a single, unified JSON tree for immediate React rendering.

POST `/api/workspace/galaxy/`

Purpose: Generates a new frame. Accepts X, Y, Width, and Height from the frontend drawing action.

POST `/api/workspace/asset/`

Purpose: Accepts FormData with a physical file and spatial coordinates, mapping the upload directly to a specific Galaxy.

PATCH `/api/workspace/asset/{id}/move/`

Purpose: Dedicated minimal endpoint triggered `onDragEnd`. Accepts only X/Y floats to persist real-time layout changes instantly.

Data Retrieval Flow

